HUMIDIFICATION

Unit Operations Lab

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INTRODUCTION

Objective:

- Study and understand the transport of both mass and heat in a wetted wall column
- Obtain the above values and compare with calculated values using theoretical equations

THEORY (DIMENSIONLESS VARIABLES)

Reynolds number

$$Re = \frac{\rho v d}{\mu}$$

Where:

- $\rho\ is\ the\ air\ density, lb/ft^3$
- $d\ is\ the\ diameter, ft$
- $\mu\ is\ the\ dynamic\ air\ viscosity, lbm/ft * s$

Schmidt number

$$Sc = \frac{V}{D_{AB}}$$

Where:

- $V\ is\ the\ kinematic\ air\ viscosity, ft^2/s$
- $D_{AB}\ is\ the\ diffustion\ coefficient\ of\ air, ft^2/s$

An example of a correlation for a gas-phase mass transfer coefficient inside a circular pipe using dimensionless groups is:

$$Sh = 0.023 * Re^{0.83} * Sc^{0.44}$$

Where:

- $Re\ is\ Reynolds\ number\ (based\ on\ pipe\ diameter)$
- $Sc\ is\ Schmidt\ number$
- $Sh\ is\ Sherwood\ number, Sh = k_c * L/D_{AB}$

THEORY

Where the mass transfer coefficient in terms of a mass ratio driving force is:

$$k_y = k_c(MW_{air})c(1-y)_M$$

Where the log mean average composition of air over the column length is:

$$(1-y)_M = \frac{\{y_1 - y_2\}}{ln\left\{\frac{(1-y)_2}{(1-y)_1}\right\}}$$

Heat transfer coefficient

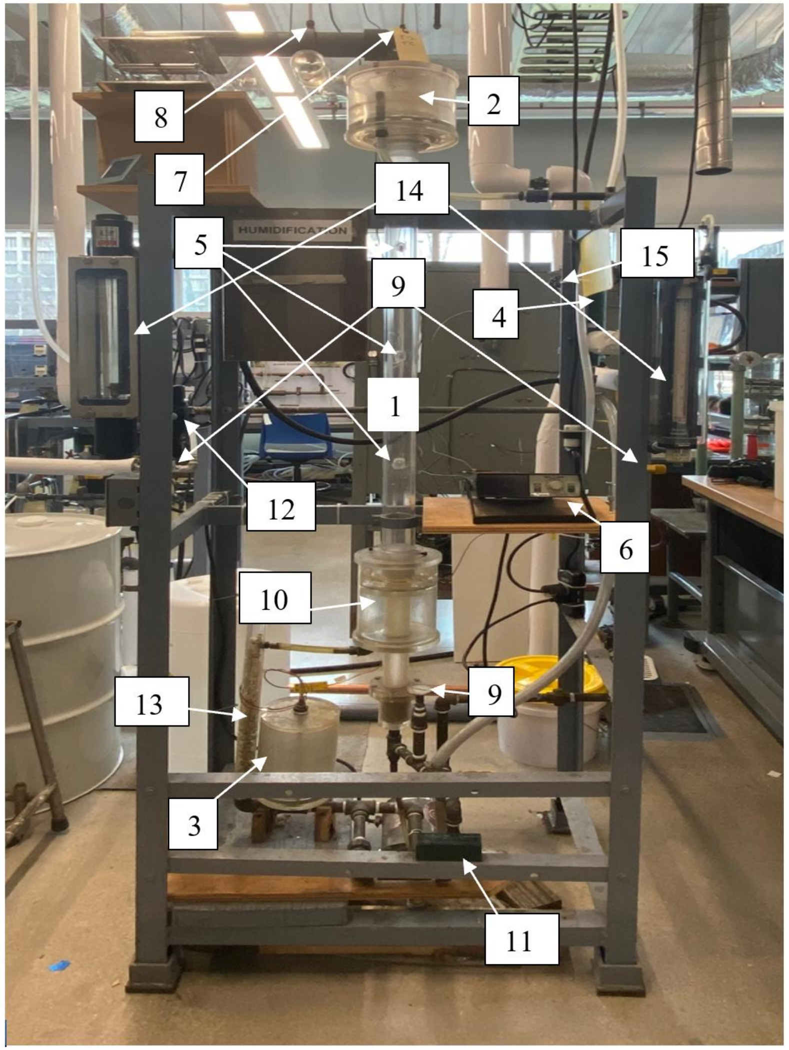
$$h = \frac{q * C_p}{Pl} \ln\left(\frac{T_i - T_1}{T_i - T_2}\right)$$

Where:

- $C_p\ is\ the\ heat\ capacity\ of\ air, Btu/ft^2 * s * °F$
- $T_i\ is\ the\ temperature\ at\ the\ interface, °F$
- $T_1\ is\ the\ inlet\ temperature, °F$
- $T_2\ is\ the\ outlet\ temperature, °F$

APPARATUS

No.	Instrument	Description
1	Humidification column	Plexiglas column in which the air and water flow through
2	Weir	Uniformly disperses water on entire surface area of column
3	Feed tank	Feeds fluid flow into the humidification column
4	Heat Exchanger	Heats air fed to the system
5	Thermocouple (5)	thermocouples at different points of the system for digital reading; measures the temperature of the water flowing down the column and the air inlet temperature
6	Digital Readout	Takes reading of all thermocouples connected to system
7	Wet bulb thermometer	Measures the temperature of the air exiting the humidifier
8	Dry bulb thermometer	Measures the temperature of the air exiting the humidifier
9	Temperature Gauge (3)	Measure the temperature of the air at the bottom of the column and two at the rotameters
10	Collection Tank	Collect fluid from column outlet
11	Pump	Centrifugal pump used to pump fluid from collection tank
12	Pressure Regulator	Regulates and provides pressure readings
13	Raschig Rings	Packing bed material
14	Rotameter (2)	One measures the air flow and the other the water flow
15	Valve	Steam control valve



MATERIALS AND SUPPLIES

Materials & Supplies	Description
Water	Fluid utilized to run flow through column
Steam	Heat supply
Compressed Air	Used to collect the condensate
Graduated Cylinder	To fill the water reservoir
Goggles	Must be worn throughout experiment for safety
Nitrile Gloves	Must be worn throughout experiment for safety
Heat Resistant Gloves	Must be worn when inserting the baffles
Hygrometer	Measures the air properties (humidity)

PROCEDURE

- Measure the humidity of incoming air using wet and dry bulb thermometers
- Record temperature across the different thermocouples for the dry run
- Circulate water in the column at a rate sufficiently enough to completely wet the inner wall of the tube
- Allow circulating water temperature to reach steady state by establishing an air flow
- After reaching steady state record the temperature across the different thermocouples and take reading from hydrometer.
- Repeat the experiment at two other flowrates
- Using the same flowrates, repeat the procedure but with two different temperatures of heated air.

EXPERIMENTAL CHALLENGES

- Determining the appropriate flowrate of steam
- Maintaining a suitable temperature in the system
- Adjusting the steam flowrate
- Attaining the same three predetermined air flowrate for all trials

SAFETY

- Avoid water spills on thermocouples, wiring and pump
- Step ladder is wobbly
- House air has a pressure of 120 psig
  - It is regulated to 40 psig in the system
- Steam has a temperature of 113°C
- Column may he hot
- Wear heat resistant gloves

QUESTIONS?

Thank You!